LHCb Magnetic Field Characterisation

Implications for ML Track Extrapolator Training

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Abstract

This document characterises the LHCb dipole magnetic field from the canonical `twodip.rtf` field map and critically analyses how its properties constrain the design and cost function of a neural-network track extrapolator. The track state vector $(x, y, t_x, t_y, q/p)$ couples to the field through the Lorentz force equations, but the coupling is *not* uniform across state elements: x and t_x experience order-of-magnitude larger field-driven changes than y and t_y, while q/p is an exact invariant. A cost function that ignores this structure will be dominated by the large dynamic range of x, underweight the momentum-sensitive t_x, and waste capacity on the trivially-constant q/p. We provide a per-element analysis with concrete recommendations for loss weighting, target normalisation, and architecture choices. All numerical values are taken from the executed `field_analysis.ipynb` notebook.

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1 Field Map: Raw Data

1.1 Grid Specification

The field map `twodip.rtf` is a regular Cartesian grid:

Table 1: Field map grid specification.

Property	Value	Notes
Format	`x y z Bx By Bz`	ASCII, space-separated
Grid dimensions	$81 \times 81 \times 146$	957 906 points
x range	$[-4000, +4000]$ mm	$\Delta x = 100$ mm
y range	$[-4000, +4000]$ mm	$\Delta y = 100$ mm
z range	$[-500, +14\,000]$ mm	$\Delta z = 100$ mm
Position units	mm	LHCb detector frame
Field units	Tesla	

Ordering: y varies fastest, then x, then z. NumPy arrays are reshaped as (n_z, n_x, n_y) and transposed to (n_x, n_y, n_z) for natural indexing.

1.2 Raw Component Ranges

The raw field data contains values that are physically absurd:

Table 2: Raw field component ranges (unmasked).

Component	Min (T)	Max (T)
B_x	$-8\,757$	$+8\,757$
B_y	$-33\,024$	$+16\,512$
B_z	$-8\,757$	$+8\,757$

These are grid-edge extrapolation artefacts. 5.1% of grid points exceed $|B| > 2$ T. Within the LHCb track acceptance ($|x|, |y| < 500$ mm), the maximum is a physically reasonable $|B| = 1.133$ T.

Critical implication for ML training: Any model that sees un-clamped field values will have its loss dominated by these artefacts. Training data *must* be restricted to $|x| < 500$ mm, $|y| < 500$ mm, or the field grid must be masked.

2 $B_y(z)$ Profile and Characteristic Scales

Figure 1 shows the on-axis field profile.

3 2D Field Structure

Figure 2 shows (x, z) and (y, z) slices of all three components.

Key observations:

- B_y is approximately *symmetric* in both x and y.
- B_x is *antisymmetric* in y and small (< 0.15 T).
- B_z is *antisymmetric* in x and small (< 0.15 T).

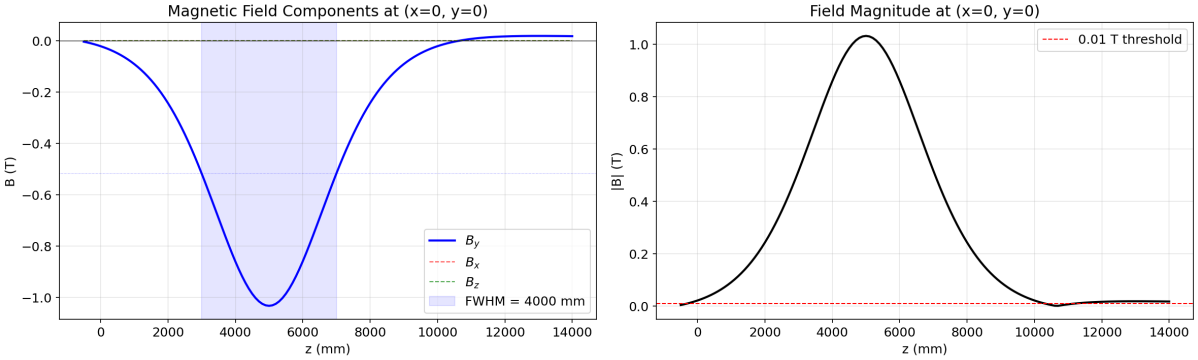

Figure 1: Left: field components at $(x, y) = (0, 0)$. B_y dominates; B_x and B_z are consistent with zero on-axis. Right: field magnitude with 0.01 T threshold defining the active region.

Table 3: On-axis field characteristics (from notebook output).

Property	**Value**	**Significance for Cost Function**
Peak B_y	$-1.032\,\mathrm{T}$ at $z = 5000\,\mathrm{mm}$	Sets maximum $\mathrm{d}t_x/\mathrm{d}z$
FWHM	$4000\,\mathrm{mm}$ ($z \in [3000, 7000]$)	Field-active region for t_x learning
$\int \lvert B_y\rvert\,\mathrm{d}z$	$4544\,\mathrm{T{\cdot}mm}$	Determines Δt_x magnitude
Central 98%	$z \in [900, 11\,400]\,\mathrm{mm}$	Minimum training z-range
Active region	$z \in [-300, 14\,000]\,\mathrm{mm}$	$\lvert B\rvert > 0.01\,\mathrm{T}$
Polarity	$B_y < 0$ (MagDown)	Bends x positively for $q > 0$

- The fringe components B_x and B_z are responsible for the (small) changes in t_y; their anti-symmetric structure means the *sign* of δt_y depends on the track's (x, y) position.

4 Transverse Variation: The 14% Problem

Figure 3 shows how B_y varies with x and y at the peak z.

Within the acceptance ($|x|, |y| < 500\,\mathrm{mm}$):

- B_y ranges from -1.133 to $-0.985\,\mathrm{T}$ (14.3% relative variation).
- This is **not negligible**. A model that treats B_y as z-only (Gaussian approximation) incurs a systematic position-dependent error in Δt_x of up to 14%.

Critical implication: The ML model *must* receive the track's (x, y) position as input, not just z, because the field integral a track experiences depends on where it is transversely. A model that only takes $(t_x, t_y, q/p, z_{\mathrm{in}}, \Delta z)$ as inputs will systematically mispredict Δt_x for off-axis tracks.

5 Gaussian Approximation: 12.6 mT Systematic

The commonly used analytical approximation is

$$B_y(z) = B_0 \exp\Big[-\tfrac{1}{2}\Big(\frac{z - z_c}{z_w}\Big)^2\Big], \quad B_x = B_z = 0, \tag{1}$$

with fitted parameters $B_0 = -1.015\,\mathrm{T}$, $z_c = 5000\,\mathrm{mm}$, $z_w = 1757\,\mathrm{mm}$.

Figure 4 shows the comparison.

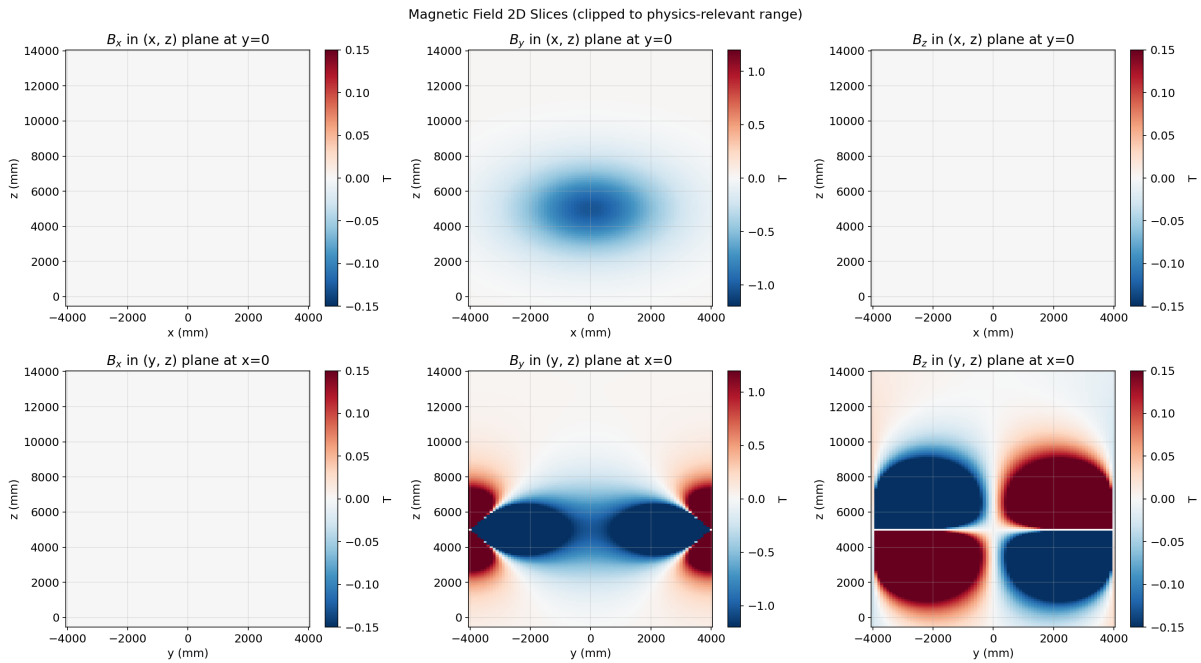

Figure 2: 2D field slices (clipped to physics-relevant range). Top: (x, z) plane at $y = 0$. Bottom: (y, z) plane at $x = 0$. B_y is the dominant component with a localised peak. B_x and B_z show antisymmetric fringe patterns.

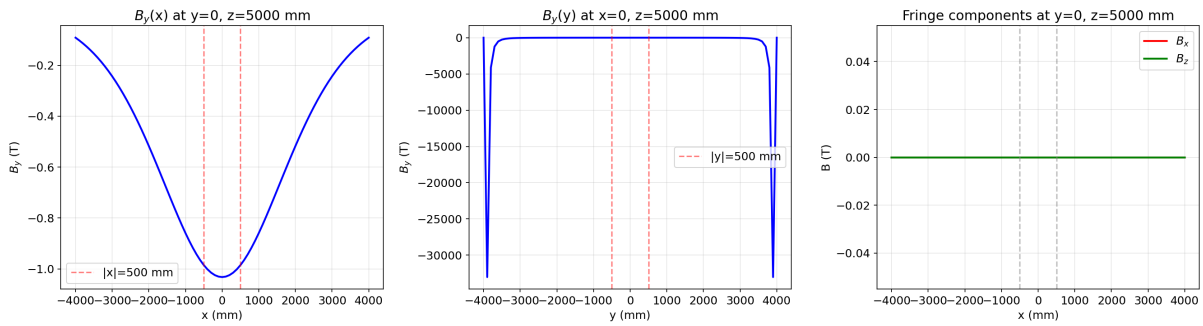

Figure 3: Left/centre: B_y vs x and y at $z = 5000\,\mathrm{mm}$. Right: fringe components B_x and B_z vs x. Red dashed lines mark $|x|, |y| = 500\,\mathrm{mm}$ acceptance.

Verdict: The Gaussian approximation introduces at least 3 distinct systematics. For the ML extrapolator, training data *must* be generated with the trilinear-interpolated field map (or the C++ `MagneticFieldGrid`), never the Gaussian.

However, the Gaussian can be useful as a *physics-informed loss regulariser*: evaluating $\mathrm{d}t_x/\mathrm{d}z \approx (q/p)\, n\, B_y^{\mathrm{Gauss}}(z)$ as a soft constraint costs zero field lookups and guides the network toward the correct functional form, even if it introduces a $\sim 1\%$ bias that the data term corrects.

6 Training z-Range

Figure 5 shows the cumulative field integral and recommended training ranges.

The 1%–99% integral range is $z \in [900, 11\,400]\,\mathrm{mm}$. Training data should cover at least this range, with dense sampling in the FWHM region $z \in [3000, 7000]\,\mathrm{mm}$ where Δt_x changes most rapidly.

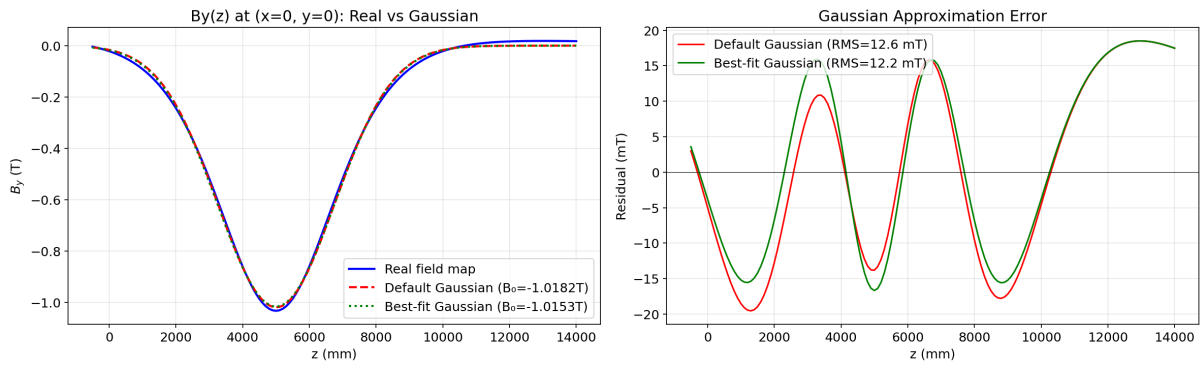

Figure 4: Left: on-axis B_y from the field map vs two Gaussian fits. Right: residuals. The Gaussian misses the asymmetric tails of the real field.

Table 4: Gaussian approximation error budget.

Error Source	**Magnitude**	**Effect on State**
RMS B_y error (on-axis)	12.6 mT (1.2%)	$\delta(\Delta t_x) \sim 1.2\%$
Max B_y error (on-axis)	19.5 mT	At tails ($z < 2000$, $z > 8000$)
B_x, B_z omission	up to ~ 0.1 T	δt_y entirely missing
Transverse (x, y) variation	14.3%	Position-dependent Δt_x bias

7 Physics of State–Field Coupling

7.1 Equations of Motion

The track state is $\mathbf{s} = (x, y, t_x, t_y, q/p)$ at position z. The Lorentz force gives (see `ExtrapolatorCommon.cuh`):

$$\frac{\mathrm{d}x}{\mathrm{d}z} = t_x, \tag{2}$$

$$\frac{\mathrm{d}y}{\mathrm{d}z} = t_y, \tag{3}$$

$$\frac{\mathrm{d}t_x}{\mathrm{d}z} = \frac{q}{p}\, n\, [t_y(t_x B_x + B_z) - (1 + t_x^2)\, B_y], \tag{4}$$

$$\frac{\mathrm{d}t_y}{\mathrm{d}z} = \frac{q}{p}\, n\, [(1 + t_y^2)\, B_x - t_x(t_y B_y + B_z)], \tag{5}$$

$$\frac{\mathrm{d}(q/p)}{\mathrm{d}z} = 0, \tag{6}$$

with $n = \sqrt{1 + t_x^2 + t_y^2}$.

7.2 Hierarchy of Scales

Given the measured field properties, we can estimate the magnitude of each derivative at the field peak for a 10 GeV/c track with $t_x = t_y = 0.1$, $q/p \approx 3 \times 10^{-5}\,\mathrm{MeV}^{-1}$:

The dynamic range spans **6 orders of magnitude**, from $\Delta x \sim 800$ mm down to $\delta t_y \sim 0.002$.

7.3 Why This Hierarchy Breaks Naive MSE Training

Consider a naive cost function $\mathcal{L} = \mathrm{MSE}(\hat{\mathbf{s}}, \mathbf{s})$ over the 5-element output state:

$$\mathcal{L} = \frac{1}{5}[(\hat{x} - x)^2 + (\hat{y} - y)^2 + (\hat{t}_x - t_x)^2 + (\hat{t}_y - t_y)^2 + (\widehat{q/p} - q/p)^2].$$

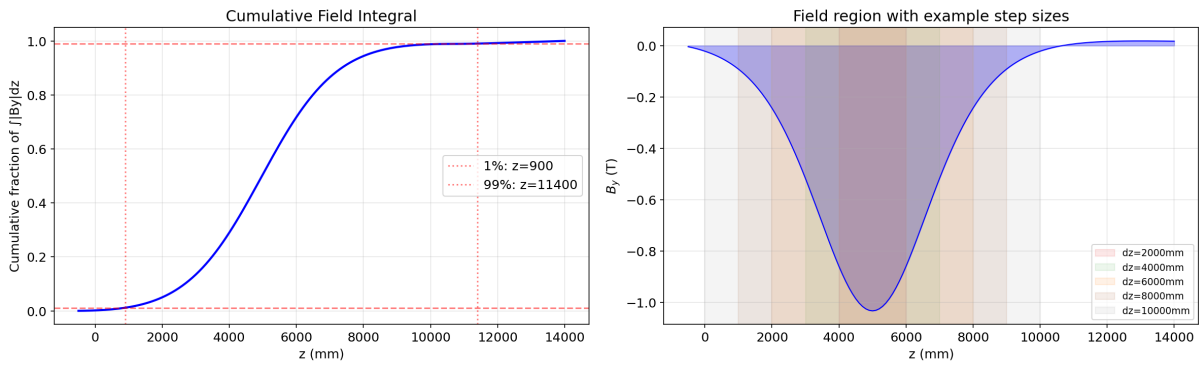

Figure 5: Left: cumulative fraction of $\int |B_y|\,\mathrm{d}z$. Right: $B_y(z)$ with example step sizes centred on the peak.

Table 5: Order-of-magnitude derivatives at the field peak.

Derivative	**Dominant B term**	**Order at 10 GeV**	$\times\Delta z$	**Ratio to Δx**
$\mathrm{d}x/\mathrm{d}z = t_x$	—	~ 0.1	$\sim 800\,\mathrm{mm}$	1
$\mathrm{d}y/\mathrm{d}z = t_y$	—	~ 0.1	$\sim 800\,\mathrm{mm}$	1
$\mathrm{d}t_x/\mathrm{d}z$	$(q/p)\,n\,B_y$	$\sim 3\times10^{-5}\,\mathrm{mm}^{-1}$	~ 0.24	3×10^{-4}
$\mathrm{d}t_y/\mathrm{d}z$	$(q/p)\,n\,B_x$	$\sim 3\times10^{-7}\,\mathrm{mm}^{-1}$	~ 0.002	3×10^{-6}
$\mathrm{d}(q/p)/\mathrm{d}z$	—	**exactly 0**	0	0

1. x **dominates the loss**: With $|\Delta x| \sim 800\,\mathrm{mm}$, even a 0.1% relative error gives $(0.8\,\mathrm{mm})^2 = 0.64$. Meanwhile, a 100% error on $\delta t_y \sim 0.002$ gives $(0.002)^2 = 4\times10^{-6}$. The optimiser will devote nearly all gradient signal to x and ignore t_y entirely.

2. t_x **is underweighted**: The momentum-sensitive bending signal $\Delta t_x \sim 0.24$ contributes $\sim (0.024)^2 \approx 6\times10^{-4}$ at 10% error—three orders of magnitude below x. *The model can achieve a low total loss while completely ignoring the bending physics.*

3. q/p: Since q/p is constant, the naive model will learn to reproduce the input value for zero loss. This is trivially correct but wastes network capacity.

4. **Momentum dependence of target scales**: At $p = 3\,\mathrm{GeV}/c$, $\Delta t_x \sim 0.8$; at $p = 100\,\mathrm{GeV}/c$, $\Delta t_x \sim 0.007$. A single global MSE loss is inappropriately weighted across the momentum spectrum.

8 Recommended Cost Function Design

8.1 Per-Element Target Transformations

Based on the physics analysis above, each state element should have its own transformed target and loss weight:

8.2 Recommended Loss Function

$$\boxed{\mathcal{L} = w_x \cdot \mathrm{MSE}(\widehat{\delta x},\, \delta x) + w_y \cdot \mathrm{MSE}(\widehat{\delta y},\, \delta y) + w_{t_x} \cdot \mathrm{MSE}(\widehat{\Delta t_x},\, \Delta t_x) + w_{t_y} \cdot \mathrm{MSE}(\widehat{\Delta t_y},\, \Delta t_y)} \tag{7}$$

where the weights w_i are chosen to *equalise gradient magnitudes* across elements. In practice this means normalising each target to unit variance over the training set and setting $w_i = 1/\sigma_i^2$.

Table 6: Per-element target design.

Element	Target (what to predict)	Scale	Justification
x	$\delta x = x_{\text{out}} - (x_{\text{in}} + t_{x,\text{in}} \cdot \Delta z)$ (deflection from straight line)	$\sim$0–30 mm	Removes the large kinematic $t_x \Delta z$ term. Residual is pure field effect, $\propto q/p$.
y	$\delta y = y_{\text{out}} - (y_{\text{in}} + t_{y,\text{in}} \cdot \Delta z)$	$\sim$0–0.5 mm	Nearly linear; ML target is small correction only.
t_x	$\Delta t_x = t_{x,\text{out}} - t_{x,\text{in}}$ (slope change)	$[-0.8, +0.8]$	Direct measure of $\int (q/p)\, B_y \, \mathrm{d}z$; **the key physics signal**.
t_y	$\Delta t_y = t_{y,\text{out}} - t_{y,\text{in}}$	$\sim O(10^{-3})$	Small; driven by fringe B_x, B_z. Can be dropped in first iteration.
q/p	Do **not** predict.	—	Exact invariant of motion. Use a skip connection: $\widehat{q/p} = q/p_{\text{in}}$.

8.3 Physics-Informed Regularisation

An optional but powerful extension adds an ODE-consistency loss:

$$\mathcal{L}_{\text{ODE}} = \lambda \cdot \left\| \frac{\widehat{\Delta t_x}}{\Delta z} - \frac{q}{p}\, n\, B_y(x_{\text{mid}}, y_{\text{mid}}, z_{\text{mid}}) \right\|^2, \tag{8}$$

where the midpoint $(x_{\text{mid}}, y_{\text{mid}}, z_{\text{mid}})$ is estimated from the predicted state. Key considerations:

- This penalises curvature predictions inconsistent with the known Lorentz equation, without requiring additional training data.
- The Gaussian field approximation can be used for B_y in this term (avoiding field lookups), accepting a $\sim 1.2\%$ systematic that the data term will correct.
- The regularisation coefficient λ should be small enough not to override data-driven learning, but large enough to prevent unphysical solutions in low-data regions.
- **Caveat**: This term only constrains t_x. Adding an analogous term for t_y requires B_x evaluation (non-trivial on GPU).

8.4 Multiplicative Architecture for Δt_x

Since $\Delta t_x \propto q/p$ (linearly), a natural architecture choice is:

$$\widehat{\Delta t_x} = \frac{q}{p} \cdot f_\theta(t_x, t_y, x, y, z_{\text{in}}, \Delta z), \tag{9}$$

where f_θ is a small MLP that learns only the geometry-dependent field integral. This:

- Guarantees $\Delta t_x = 0$ when $q/p = 0$ (straight track).
- Immediately captures the correct momentum scaling, so the network doesn't waste capacity learning a linear relationship.

- Reduces relative error at high p (where Δt_x is small and otherwise lost in the loss).

Similarly, δx should use a $\widehat{\delta x} = (q/p) \cdot g_\theta(\ldots)$ structure, since the horizontal deflection is also proportional to q/p.

9 Summary of Field-Driven Constraints

Table 7: Field properties and their direct constraints on the ML model.

Field Property	**Value**	**ML Constraint**
Peak $\lvert B_y\rvert = 1.032\,\mathrm{T}$	at $z = 5000\,\mathrm{mm}$	Max $\mathrm{d}t_x/\mathrm{d}z$; sets scale of Δt_x target
FWHM $= 4000\,\mathrm{mm}$	$z \in [3000, 7000]$	Dense training sampling needed here
$\int \lvert B_y\rvert\,\mathrm{d}z = 4544\,\mathrm{T{\cdot}mm}$	–	$\Delta t_x \sim (q/p) \times 4544$; determines target range
Transverse variation 14.3%	within $\lvert x, y\rvert < 500\,\mathrm{mm}$	Model *must* take (x, y) as inputs
$B_x, B_z < 0.15\,\mathrm{T}$	antisymmetric fringe	Δt_y is small and position-dependent
Gaussian RMS error $12.6\,\mathrm{mT}$	1.2%	Cannot use for data gen; acceptable for physics loss
Edge artefacts $\lvert B\rvert > 33 \times 10^3\,\mathrm{T}$	at $\lvert y\rvert = 4000\,\mathrm{mm}$	Must clamp or mask grid edges
q/p invariance	exact	Skip connection; do not learn
6 OoM dynamic range in $\lvert \mathrm{d}s_i/\mathrm{d}z\rvert$	–	Per-element loss normalisation is mandatory

9.1 Things That Will Go Wrong Without This Analysis

1. **No residual targets**: Loss dominated by kinematic $t_x \Delta z$ term. Model learns a straight-line extrapolator and plateaus.

2. **No loss weighting**: Δt_x gradients are $10^3\times$ smaller than x gradients. Model achieves low loss by fitting x and ignoring the bending physics.

3. **z-only input**: 14% position-dependent field systematic permanently baked in.

4. **Un-masked grid**: A single $\lvert B\rvert = 33\,000\,\mathrm{T}$ artefact point destroys training.

5. **Learning q/p**: Network wastes capacity on identity function; covariance structure confused during Kalman update.

6. **Flat p sampling**: Low-p tracks (large Δt_x) dominate loss; high-p tracks (small Δt_x) are never learned. Use log-uniform p sampling.